VIGNESH T

Senior Robotics Engineer · Autonomous Systems · ROS2 / Nav2 / SLAM / Edge AI

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Senior Robotics Engineer with 6+ years of experience building and deploying robotic systems across defense, logistics, hospitality, and outdoor environments. Deep expertise in ROS2, Nav2, SLAM, GPS-denied localization, Visual-Inertial Odometry (VIO), and edge AI deployment on NVIDIA Jetson platforms. Proven record of owning full robotics infrastructure end-to-end — from hardware integration through simulation, testing, and production deployment — and leading engineering teams across complex, multi-robot programs.

CORE TECHNICAL SKILLS

Autonomy & Navigation: ROS2, Nav2, SLAM, Motion Planning, Sensor Fusion, GPS-Denied Localization, Visual-Inertial Odometry (VIO), Visual Localization, Behavior Trees, Collision Avoidance, Multi-Robot Coordination, OpenRMF, Fleet Management

Perception & Edge AI: LiDAR, RGB-D Cameras, Mono & Thermal Cameras, IMU, Point Cloud Processing (PCL), NVIDIA Jetson Orin, Edge AI Inference, Optimization for Constrained Platforms

Simulation & Infrastructure: NVIDIA Isaac Sim, Gazebo, Unity, Sensor Simulation, Scenario Generation, Hardware-in-the-Loop (HIL) Testing, CI/CD, Regression Testing

Programming & Tools: C++, Python, ROS2 Control, Docker, Linux, Git, MQTT, REST APIs, URDF, RViz

Robot Platforms: Differential Drive, Ackermann, Holonomic (Mecanum), Swerve Drive, Tracked, Quadruped, Unmanned Aerial Vehicles (UAVs)

AI Robotics: PyTorch, Edge AI Inference, Imitation Learning (learning), Reinforcement Learning (learning)

PROFESSIONAL EXPERIENCE

Senior Robotics Engineer · **Paar Autonomy** · Bangalore, India *Oct 2025 – Present*

- Developing AI-driven drone platforms for defense applications, focusing on GPS-denied navigation, real-time perception, and mission-critical decision-making in GNSS-contested environments.
- Built large-scale simulation pipelines in NVIDIA Isaac Sim for sensor simulation, scenario generation, and automated regression testing across 50+ defense mission scenarios.
- Designed GPS-denied geo-position estimation using Visual-Inertial Odometry (VIO), monocular feature matching, and visual localization against satellite imagery.
- Optimized perception and localization pipelines for real-time edge AI inference on NVIDIA Platform, improving throughput for deployment on weight-constrained drone platforms.
- Integrated and calibrated mono, EO, thermal, and GNSS sensor arrays for end-to-end navigation workflows on unmanned systems.

Robotics Engineer · **Kabam Robotics** · Singapore *Nov 2024 – Oct 2025*

- Designed LiDAR and RGB-D perception modules; developed SLAM-based navigation for GPS-denied security patrol across large outdoor environments.
- Contributed Nav2 map server optimization — merged upstream — improving large-map loading performance for deployments spanning 10,000+ m².
- Built a production-grade multi-map management package enabling scalable navigation across multi-zone facilities.
- Implemented behavior tree–based decision-making for patrol operations; enhanced Robot Health Management Systems for real-time remote monitoring.

Research Engineer – Motion Planning · **Novus Hi-Tech** · Bangalore, India*Apr 2024 – Nov 2024*

- Developed motion planning and trajectory optimization for swerve drive, tracked, quadruped, and Ackermann robotic platforms operating in unstructured defense terrain.
- Engineered full-stack kinematic control using forward/inverse kinematics integrated with Nav2; contributed to navigation stacks for all-terrain unmanned ground vehicles.
- Collaborated with DRDO scientists and IIT/NIT researchers on multi-domain platform research; built HIL simulation environments in Unity for field validation.

Robotics Software Engineer · Team Lead · **Goat Robotics (formerly Katomaran Technologies)** · Coimbatore, India*2020 – Apr 2024*

- Architected and deployed end-to-end AMR systems using ROS2 across hospitality, warehousing, and logistics verticals; led a 5-member cross-functional engineering team.
- Owned complete robotics infrastructure — SLAM, localization, sensor fusion, navigation, perception, real-time decision-making — delivering 6+ production robotic platforms.
- Integrated OpenRMF fleet management with lift/door control, task bidding, and traffic management for large-scale multi-robot enterprise deployments.
- Designed docking systems using AprilTag detection, point cloud alignment, and odometry-based control — achieving centimeter-level repeatability across all charging stations.
- Built cloud-connected fleet management with REST APIs, MQTT telemetry, Android HMI, and OTA update pipelines; implemented CI/CD with automated regression testing.

SELECTED PROJECTS

GPS-Denied Drone Localization · Defense

- Fused VIO, monocular feature matching, and satellite-image-based visual localization for drone platforms in GNSS-contested environments.

LiDAR-Based Person Following Robot · Robotics / Autonomous Navigation

- Developed a ROS2-based person-following robot using a machine learning package trained on human leg patterns for detection using only 2D LiDAR data.
- Designed custom tracking and control algorithms to identify, lock, and follow selected target IDs in real time for autonomous human-following applications.

Autonomous Hotel Delivery Robot · Hospitality

- Full-stack AMR with vending system integration, OTP-based delivery verification, precision docking, and Android HMI; sustained 100+ daily deliveries with >99% task completion in live hotel deployments.

Autonomous Trolley Detection & Docking System · Warehouse Robotics

- Developed an ICP-based point cloud matching system using Intel RealSense depth camera data and PCL for autonomous trolley detection in warehouse material-handling environments.
- Implemented docking, lifting, and dropping control algorithms by matching live point cloud data against preloaded trolley PCD models for precise autonomous material transfer operations.

LEADERSHIP & OPEN SOURCE

- Led 5-member robotics engineering team across robotics infrastructure development, integration, sprint planning, code review, and release management.
- Active contributor to the Nav2 open-source project — large-map loading optimization merged upstream.
- Collaborated with DRDO scientists, IIT/NIT researchers, and international robotics developers on defense unmanned systems programs.

EDUCATION

B.E. Mechatronics · Sri Krishna College of Engineering and Technology, Tamil Nadu · 2016 – 2020